

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022**(CBCSS-UG)****Mathematics****MTS 6P 15—RESEARCH METHODOLOGY****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6P 15—RESEARCH METHODOLOGY
(Multiple Choice Questions for SDE Candidates)

1. Choose the false statement.

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|---|---|
| (A) Inclusion ($\subset$) is reflexive. | (B) Belonging ($\in$) is not reflexive. |
| (C) Inclusion ($\subset$) is symmetric. | (D) Belonging ($\in$) is not symmetric. |

2. Which of the following is true ?

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|---|-------------------------------|
| (A) $\{\phi\} = \phi$. | (B) $\phi \in \{\{\phi\}\}$. |
| (C) $\{\{\phi\}\} = \{\{\phi\}, \{\phi\}\}$. | (D) $\{\phi\} \subset \phi$. |

3. "Pure mathematics is, in its way, the poetry of logical ideas." is said by :

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|----------------------|--------------------|
| (A) Plato. | (B) Mark Twain. |
| (C) Albert Einstein. | (D) Thomas Storer. |

4. PCTeX is related to :

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|------------------|-------------|
| (A) Mathematica. | (B) Maple. |
| (C) Latex. | (D) Octave. |

5. The word research is derived from the French word :

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|---------------|--------------|
| (A) Research. | (B) Recerch. |
| (C) Resourch. | (D) Riserch. |

6. Research is related with :

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|---------------------------------|----------------------------|
| (A) Discovery of new idea. | (B) Solution of a problem. |
| (C) Investigation of a problem. | (D) All of the above. |

7. Which among the following can be taken as the discrete object ?

- | | |
|---------------|---------------------------|
| (A) People. | (B) Rational numbers. |
| (C) Integers. | (D) All of the mentioned. |

8. Which is correct syntax for $\text{\LaTeX}$ Table ?

- (A) $\text{\textbackslash begin}{tables}.$
- (B) $\text{\textbackslash begin}{table}.$
- (C) $\text{\textbackslash begin}{tabular}.$
- (D) $\text{\textbackslash begin}{tbl}.$

9. Syntax for $\supset$.

- (A) $\text{\textbackslash subset}.$
- (B) $\text{\textbackslash superset}.$
- (C) $\text{\textbackslash supset}.$
- (D) None of these.

10. File extension of a $\text{\LaTeX}$ file :

- (A) .tex
- (B) .doc.
- (C) .latex
- (D) None of these.

11. From a practical perspective, the first thing you need to do before designing an algorithm is _____.

- (A) Understanding the Problem.
- (B) Analyzing an algorithm.
- (C) Coding an algorithm.
- (D) None of these.

12. Once an algorithm has been specified, you have to prove its _____.

- (A) Specification.
- (B) Necessity.
- (C) Output.
- (D) Correctness.

13. Name the problem that is to rearrange the items of a given list in nondecreasing order.

- (A) Searching problem.
- (B) Graph problem.
- (C) Sorting problem.
- (D) Numerical problem.

14. How to align mathematical formula in center ?

- (A) $\text{\textbackslash $... $}.$
- (B) $\text{\textbackslash $$ $$}.$
- (C) $\text{\textbackslash $ $$}.$
- (D) None of these.

15. What is the output of $\text{\textbackslash Rightarrow}$?

- (A) $\Rightarrow.$
- (B) $\rightarrow.$
- (C) $\Leftrightarrow.$
- (D) $\leftrightarrow.$

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6P 15—RESEARCH METHODOLOGY

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. State the axiom of extension. What does the axiom say if $A = \{1, 2\}$ and $B = \{1, 2, 3\}$?
2. Why is the empty set subset of every set ?
3. State Demorgan's laws.
4. Define a transitive relation. Give an example.
5. For a function $f : X \rightarrow Y$, define f^{-1} . If $X = \{-1, 1\}$, $Y = \{-1, 1\}$, $f(x) = x^2$, what is $f^{-1}(-1)$?
6. Give two reasons for learning mathematics.
7. What are two types of mathematical proofs ?
8. What is meant by an algorithm ? Give an example (without details) where an algorithm can be devised.
9. Explain the terms approximation algorithm and exact algorithm.
10. Write the $\text{\LaTeX}$ code for producing the two sentences :
Hello world !
How are you ?
11. Write the $\text{\LaTeX}$ code for producing the symbols $\in, \subset, \neq, \Rightarrow$.
12. Name two popular Linux distributions.

(8 × 3 = 24 marks)

Turn over

Section B

*Answer at least five questions.
Each question carries 5 marks.
All questions can be attended.
Overall Ceiling 25.*

13. Show that (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$; (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$; (iii) $A \cap \phi = \phi$.
14. Show that (i) $(A \cup B) \times X = (A \times X) \cup (B \times X)$; (ii) $(A - B) \times X = (A \times X) - (B \times X)$.
15. Give a proof by mathematical induction for the statement: " $5n + 6 \cdot 7^n + 1$ is divisible by 8 for all non-negative integers n ".
16. What are the techniques that are a central part of the mathematical thought process?
17. Explain the consecutive integer checking algorithm for computing $\gcd(m, n)$.
18. Give a diagrammatic representation of the design and analysis process of an algorithm.
19. Write a complete $\text{\LaTeX}$ document to produce the following output given in the box :

1. Let $n = 3$. Then $n^2 + 1 = 10$.
2. The curve $y = \sqrt{x}$, where $x \geq 0$, is concave downward.
3. If $\sin \theta = 0$ and $0 \leq \theta < 2\pi$, then $\theta = 0$ or $\theta = \pi$.

(5 × 5 = 25 marks)

Section C

*Answer any one question.
The question carries 11 marks.*

20. Write a note on preparing a mathematical talk giving details on :
 - (i) do's and don'ts while giving a talk.
 - (ii) most important resources available for presenting mathematics.
21. (i) Explain any two problem types in algorithm.
- (ii) Write a short note on two of the differences in storing files in Linux and Windows.

(1 × 11 = 11 marks)